

Bridging the *real* and *imaginary* worlds

Le Chen

Auburn University

Stochastic Analysis Seminar

Auburn University

2022-04-19

Plan

A bird view of SPDEs and the two worlds

Fractional PDE – the tool to bridge the two worlds

One case study of SPDEs in two worlds

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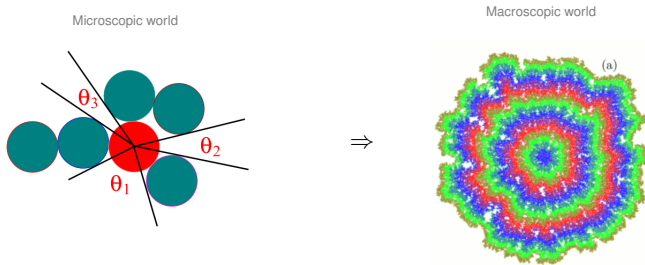
Fractional PDE – the tool to bridge the two worlds

One case study of SPDEs in two worlds

Surface growth

Rule of replication of **cells**¹

Replication probability $\propto$ Aperture angle θ_i



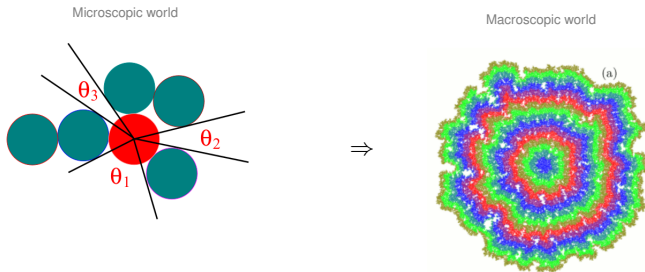
$$\frac{\partial}{\partial t} h(t, x) = \frac{1}{2} \Delta h(t, x) + \frac{\lambda}{2} (\nabla h)^2 + \dot{W}(t, x) \quad (\text{KPZ})$$

¹S. Santalla and S. Ferreira. *Phys. Rev. E* 98, 2018

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Some rules

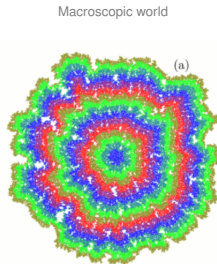
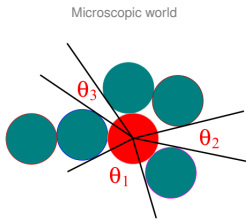
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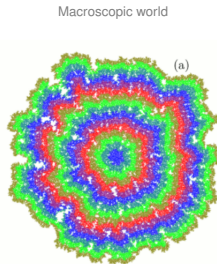
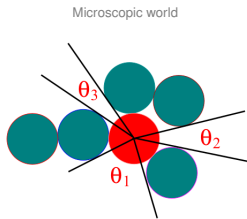
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Study of growing interfaces in a thin film

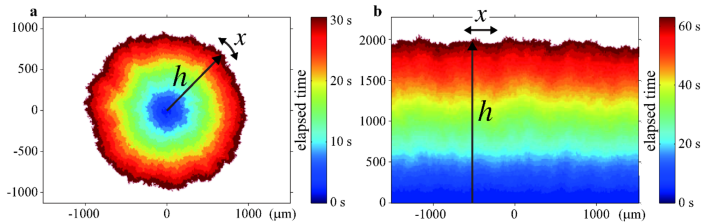
— Convection of nematic liquid crystal ²

Show movies !

²K. Takeuchi, M. Sano, T. Sasamoto *et al.* *Sci. Rep. (Nature)*, 1, 34 (2011).

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Prediction from KPZ equation:

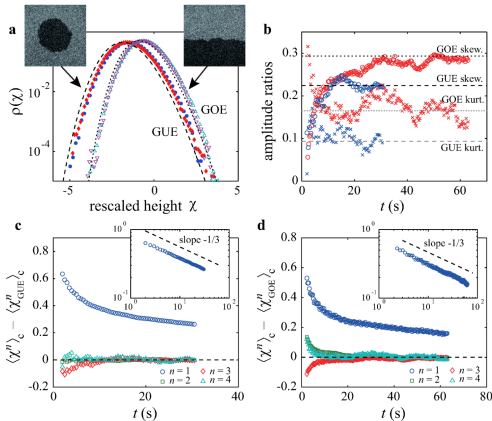
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KPZ Equation '86 ³



Mehran Kardar (1957 –) Giorgio Parisi (1948 –)

Yicheng Zhang

³"Dynamic Scaling of Growing Interfaces". Physical Review Letters. 56 (9): 889–892, 1986.

Giorgio Parisi

Facts



III. Niklas Elmehed © Nobel Prize Outreach

Giorgio Parisi
The Nobel Prize in Physics 2021

Born: 4 August 1948, Rome, Italy

Affiliation at the time of the award: Sapienza University of Rome, Rome, Italy

Prize motivation: "for the discovery of the interplay of disorder and fluctuations in physical systems from atomic to planetary scales."

Prize share: 1/2

<https://www.nobelprize.org/prizes/physics/2021/parisi/facts/>

KBE FRS



Website hairer.org

- ▶ **M. Hairer**. Solving the KPZ equation. *Annals of Mathematics*, vol. 178, 2013.
- ▶ **M. Hairer**. A theory of regularity structures. *Inventiones mathematicae*, vol. 198, 2014.

Fields Medal in 2014

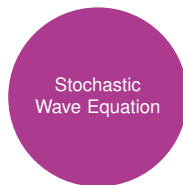
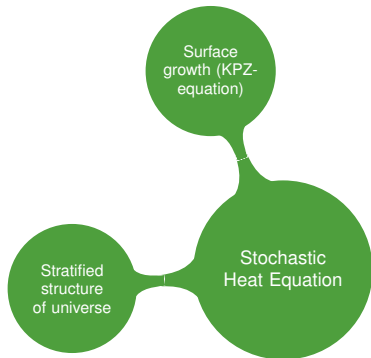
Stochastic
Heat Equation

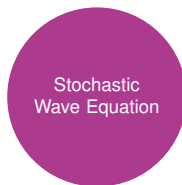
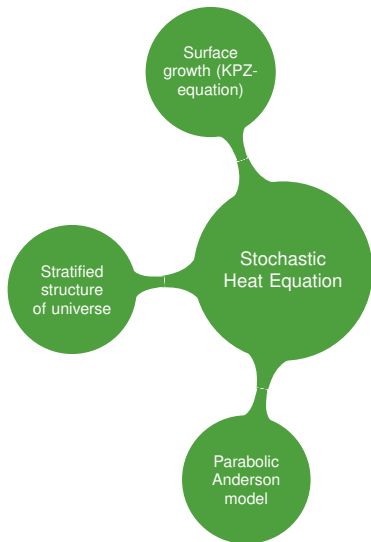
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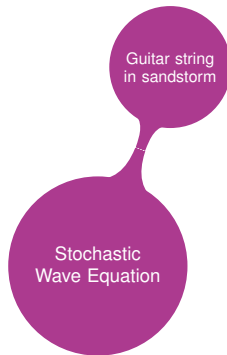
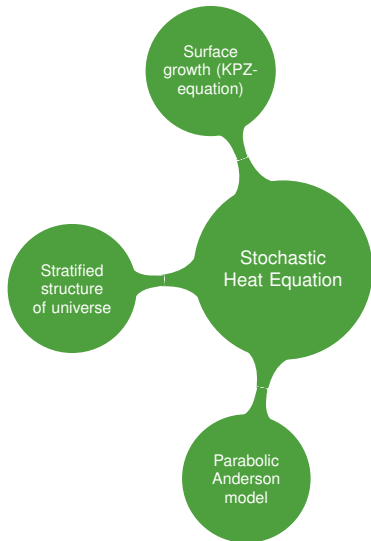
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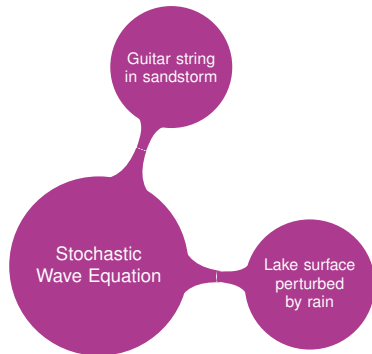
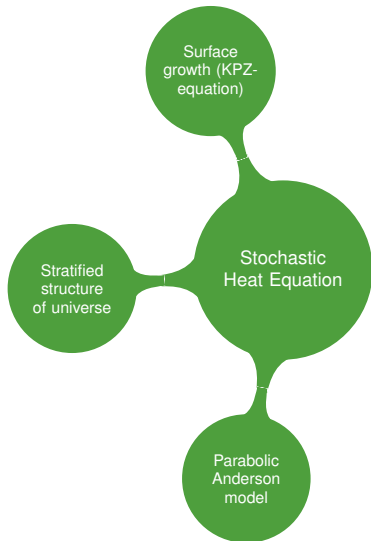
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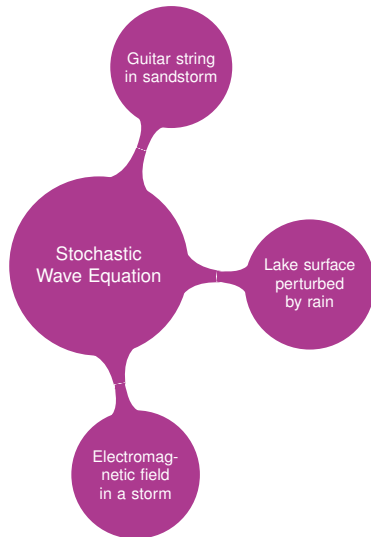
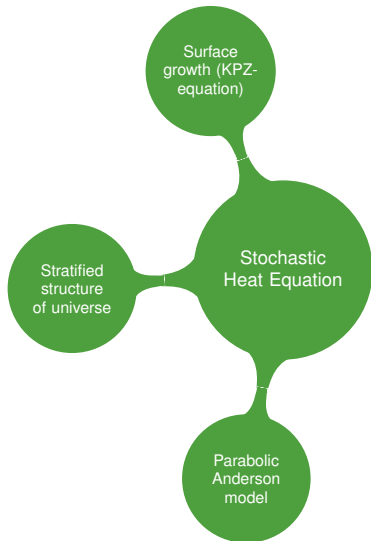
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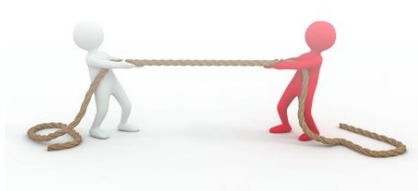




$$\frac{\partial^b}{\partial t^b} u(t, x) = \left(\frac{1}{2} \Delta + \dot{W}(x) \right) u(t, x)$$

$b = 1$: SHE

$b = 2$: SWE



Δu

$\dot{W}u$

Smoothing

Roughening

$$\frac{\partial}{\partial t} u(t, x) = \left(\frac{1}{2} \frac{\partial^2}{\partial x^2} + \lambda \dot{W}(t, x) \right) u(t, x)$$

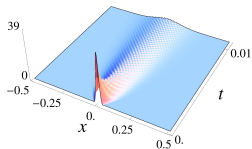
— Propagation of tall peaks ⁴

⁴C. & Dalang, *Ann. of Prob.*, 43 (6), (2015).

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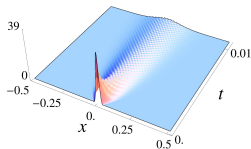


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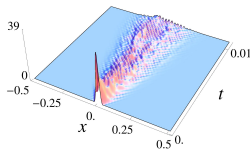
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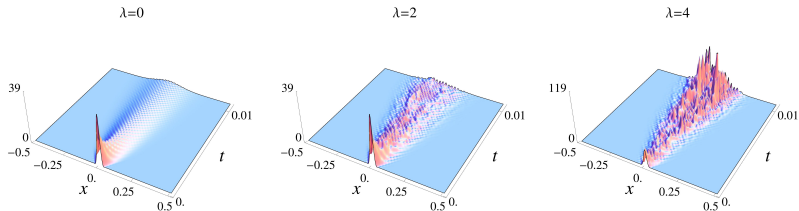
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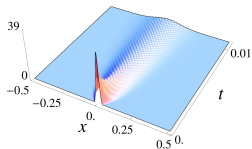


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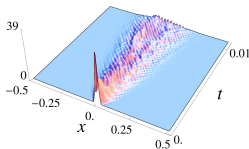
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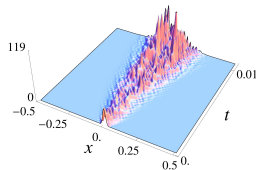
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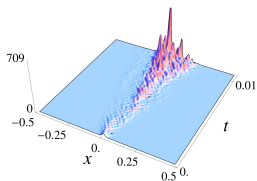
$\lambda=2$



$\lambda=4$



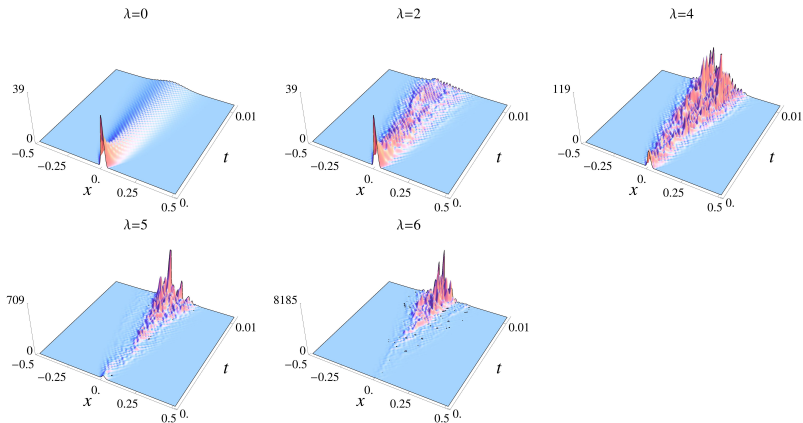
$\lambda=5$



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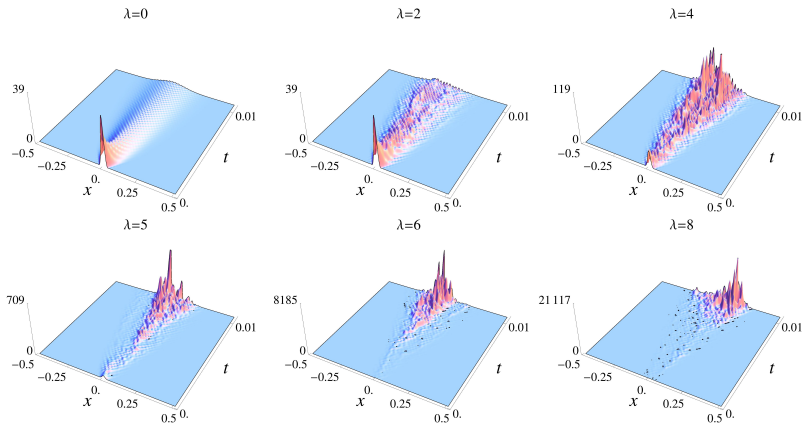
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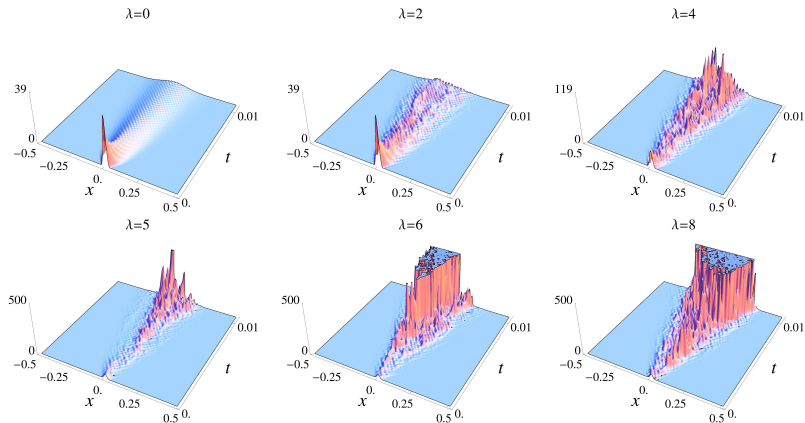
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$$G_H(t, x) = (2\pi t)^{-d/2} \exp\left(-\frac{\|x\|^2}{2t}\right) \quad t > 0, x \in \mathbb{R}^d$$

— v.s. —

$$G_W(t, x) = \begin{cases} \frac{1}{2} 1_{\{|x| < t\}} & \text{if } d = 1 \\ \frac{1}{2\pi} \frac{1}{\sqrt{t^2 - |x|^2}} 1_{\{|x| < t\}} & \text{if } d = 2 \\ \frac{1}{4\pi t} \sigma_t & \text{if } d = 3 \\ C_d \left[\left(t^{-1} \partial_t \right)^{\frac{d-3}{2}} \left(t^{d-2} \int_{|y|=1} \delta_0(x + ty) d\sigma(y) \right) \right] & \text{if } d \geq 3 \text{ odd} \\ C'_d \left[\left(t^{-1} \partial_t \right)^{\frac{d-2}{2}} \left(t^{d-1} \int_{|y|=1} \frac{\delta_0(x + ty)}{\sqrt{1 - |y|^2}} dy \right) \right] & \text{if } d \geq 2 \text{ even} \end{cases}$$

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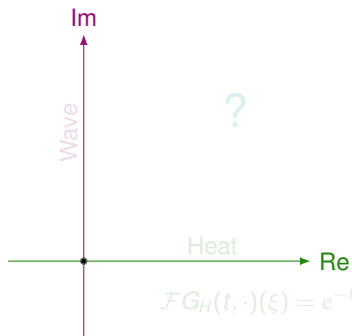
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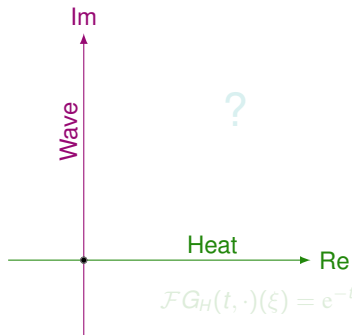


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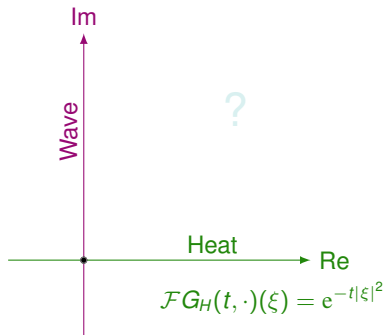


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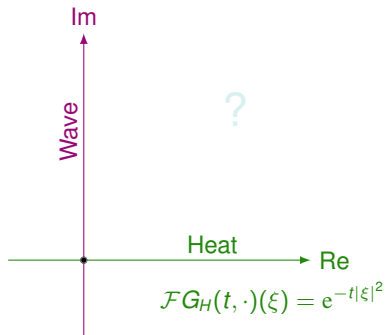
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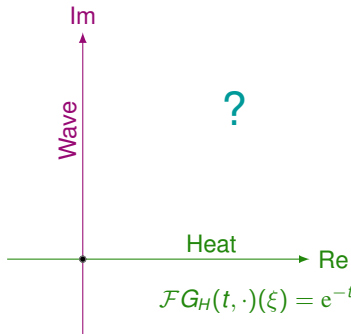
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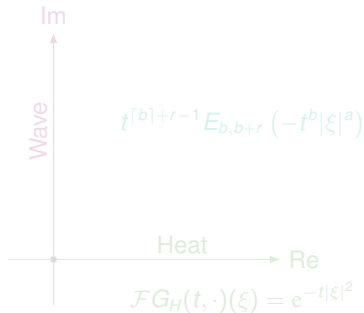
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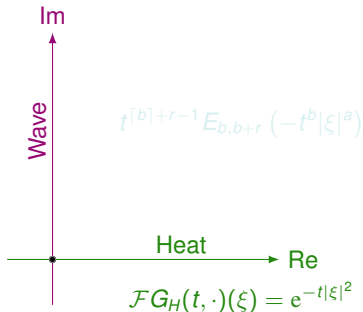
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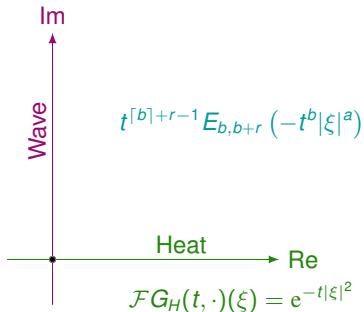
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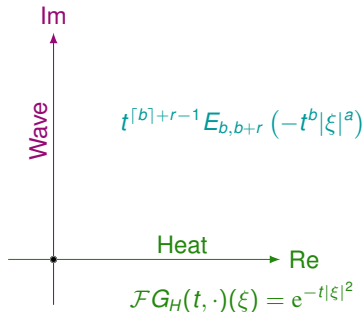
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$$\left(\partial_t^b + (-\Delta)^{a/2}\right) u(t, x) = I_t^r f(t, x), \quad u(t, x) = \int_0^t \int_{\mathbb{R}^d} G_{a,b,r,d}(t-s, x-y) f(s, y) ds dy$$

$$\mathcal{F}G_W(t, \cdot)(\xi) = \frac{\sin(t|\xi|)}{|\xi|} = \operatorname{Im} \frac{e^{it|\xi|}}{|\xi|}$$

$$\mathcal{L}\mathcal{F}G_W(\circ, \cdot)(s, \xi) = \frac{1}{s^2 + |\xi|^2}$$



$$\frac{s^{-r}}{s^b + |\xi|^a}$$

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$$\mathcal{LF}G_{a,b,r,d}(\circ, \cdot)(s, \xi) = \frac{s^{-r}}{s^b + |\xi|^a}$$

$\Downarrow$

$$\mathcal{FG}_{a,b,r,d}(t, \cdot)(\xi) = t^{\lceil b \rceil + r - 1} E_{b,b+r} \left(-t^b |\xi|^a \right)$$

$\Downarrow$

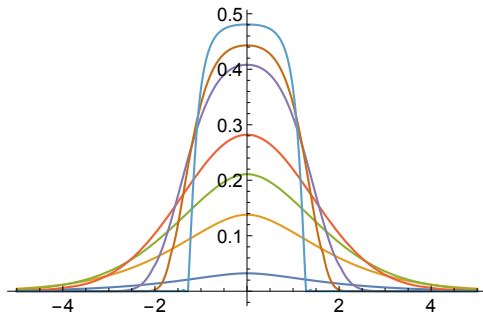
$$G_{a,b,r,d}(t, x) = \pi^{-d/2} |x|^{-d} t^{b+r-1} H_{2,3}^{2,1} \left(\frac{|x|^a}{2^{a-1} \nu t^b} \middle| \begin{matrix} (1,1), (b+r,b) \\ (d/2, a/2), (1,1), (1, a/2) \end{matrix} \right)$$

$G_{2,\beta,0,1}(1, x)$ with ⁵

$\beta = (2,) \quad 15/8, \quad 5/3, \quad 3/2, \quad 1, \quad 3/4, \quad 1/2, \quad 1/8$

$\uparrow$
 $\frac{1}{2} 1_{[-1,1]}(x)$

$\uparrow$
 $\frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$



$G_{2,\beta,0,1}(t, x)$ with ⁶

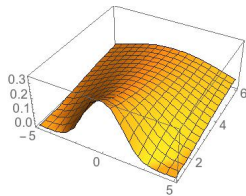
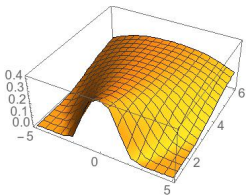
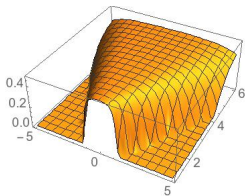
$$\beta = (2,) \quad 15/8, \quad 3/2, \quad 6/5, \quad (1)$$

$$\uparrow$$

$$\frac{1}{2} 1_{[-t,t]}(x)$$

$$\uparrow$$

$$\frac{1}{\sqrt{2\pi t}} e^{-\frac{x^2}{2t}}$$



$$(t, x) \in [1, 6] \times [-5, 5]$$

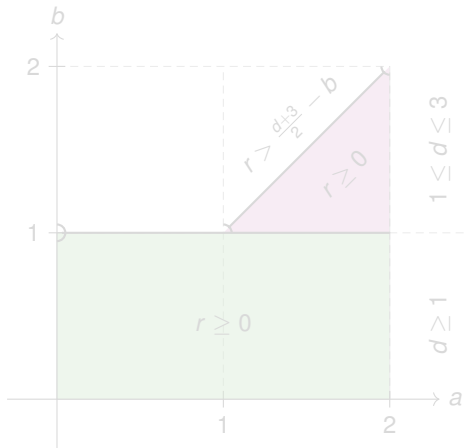
$$A_a = -\pi^{-d/2} \nu 2^{a-1} \frac{\Gamma((d+a)/2)}{\Gamma(2b+r)\Gamma(-a/2)}$$

$$G_{a,b,r,d}(1,x) \sim \begin{cases} A_a |x|^{-(d+a)} & \text{if } a \neq 2 \\ A_2 |x|^\alpha \exp(-\beta |x|^c) & \text{if } a = 2 \end{cases} \quad \text{as } |x| \rightarrow \infty$$

$$A_2 = \pi^{-d/2} (2-b)^{d/2-(b+r)} b^{\frac{b(d+2-2(b+r))}{2(2-b)}} (2\nu)^{\frac{2(b+r)-(d+2)}{2(2-b)}}$$

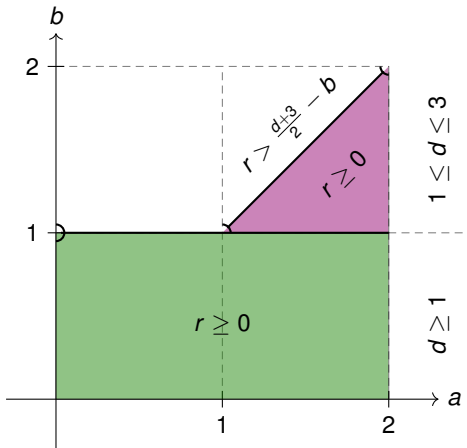
$$\alpha = \frac{d(b-1) - 2(b+r-1)}{2-b}, \quad \beta = (2-b)b^{\frac{b}{2-b}} (2\nu)^{\frac{1}{b-2}}, \quad c = \frac{2}{2-b}.$$

When is $G_{a,b,r,d}(t, x)$ nonnegative?



C., Hu, Nualart, 19

When is $G_{a,b,r,d}(t, x)$ nonnegative?



C., Hu, Nualart, 19

Plan

A bird view of SPDEs and the two worlds

Fractional PDE – the tool to bridge the two worlds

One case study of SPDEs in two worlds

$$\mathbb{E}[W(\varphi)W(\psi)] = \int_{\mathbb{R}^d} \mathcal{F}\varphi(\xi) \overline{\mathcal{F}\psi(\xi)} \mu(d\xi) = (2\pi)^d \iint_{\mathbb{R}^{2d}} \varphi(x) \psi(y) \gamma(x-y) dx dy$$

spectral measure

correlation function

nonnegative $\iff$ nonnegative definite

nonnegative definite $\iff$ nonnegative

$\mu = \varphi(\xi)$	$\gamma(x)$
$C_{d,\alpha} \xi ^{-(d-\alpha)}$	$ x ^{-\alpha}$
$(2\pi)^{-d}$	δ_0

Assumptions	
A: $\alpha \in (0, d)$	
B: $d = 1, 2$	C: $d = 3$

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Assumptions	
A: $\alpha \in (0, d)$	
B: $d = 1, 2$	C: $d = 3$

$$\begin{cases} \frac{\partial^2 u}{\partial t^2}(t, x) &= \Delta u(t, x) + \sqrt{\theta} u(t, x) \dot{W}(x), \quad t > 0, x \in \mathbb{R}^d \\ u(0, x) &= 1, \quad \frac{\partial u}{\partial t}(0, x) = 0 \end{cases}$$

- Time-independent centered Gaussian noise:

$$\mathbb{E}[W(\varphi)W(\psi)] = \int_{\mathbb{R}^d} \mathcal{F}\varphi(\xi) \overline{\mathcal{F}\psi(\xi)} \mu(d\xi) =: \langle \varphi, \psi \rangle_{\mathcal{H}}$$

Question:

- Well-posedness
- Exact large-time/moment asymptotics

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Mild solution (in Skorohod sense)

$$u(t, x) = 1 + \sqrt{\theta} \int_0^t \left(\int_{\mathbb{R}^d} G(t-s, x-y) u(s, y) W(\delta y) \right) ds$$

Solved through

$$u(t, x) = 1 + \sum_{n=1}^{\infty} \theta^{n/2} I_n(f_n(\cdot, x, t))$$

where

$$f_n(x_1, \dots, x_k; x, t) = \int_0^t \int_0^{t_n} \cdots \int_0^{t_2} G(t-t_n, x-x_n) \cdots G(t_2-t_1, x_2-x_1) dt_1 \cdots dt_n$$

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It is well known that⁷

$L^2(\Omega)$ solution exists on $(0, T) \times \mathbb{R}^d$



$$\mathbb{E} \left(u(t, x)^2 \right) = \sum_{n \geq 0} \theta^n n! \left\| \tilde{f}_n(\cdot, x, t) \right\|_{\mathcal{H}^{\otimes n}}^2 < \infty \quad \forall (t, x) \in (0, T) \times \mathbb{R}^d$$

⁷ $\tilde{f}_n(x_1, \dots, x_k; x, t) = \frac{1}{n!} \sum_{\rho \in \Sigma_n} f_n(x_{\rho(1)}, \dots, x_{\rho(n)})$

How to get sharp estimate of $\left\| \widetilde{f}_n(\cdot, x, t) \right\|_{\mathcal{H}^{\otimes n}}^2$?

By scaling property:

$$\left\| \widetilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 = \left\| \widetilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 t^{\left[2(b+r) - \frac{b\alpha}{a}\right]n}$$

so we only need to show how fast the following function decays in n :

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How to get sharp estimate of $\left\| \tilde{f}_n(\cdot, x, t) \right\|_{\mathcal{H}^{\otimes n}}^2$?

We want to use Laplace transform to probe the growth in t in terms of n :

$$n \mapsto \int_0^\infty e^{-t} \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt$$

E.g. $\int_0^\infty e^{-t} t^n dt = 1/(\Gamma(n+1))$ for all $n > -1$.

$$\int_0^\infty e^{-t} e^{\sqrt{t}} dt = \frac{1}{2} \sqrt[4]{e} \sqrt{\pi} (\operatorname{erf}(1/2) + 1) + 1 \approx 2.730.$$

$$\int_0^\infty e^{-t} e^{-\sqrt{t}} dt = 1 - \frac{1}{2} \sqrt[4]{e} \sqrt{\pi} \operatorname{erfc}(1/2) \approx 0.454.$$

$$\int_0^\infty e^{-t} \sin(t) dt = \int_0^\infty e^{-t} \cos(t) dt = 1/2.$$

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Diagonalization: Let $\gamma = K * K$ and define

$$H_n(t, \vec{x}) = n! \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n K(x_k - y_k) \tilde{f}_n(\vec{y}; 0, t) d\vec{y}$$

so that

$$\left\| \tilde{f}_n(\cdot, 0; t) \right\|_{\mathcal{H}^{\otimes n}}^2 = \frac{1}{(n!)^2} \int_{(\mathbb{R}^d)^n} H_n^2(t, \vec{x}) d\vec{x}$$

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\end{aligned}$$

where

$$\frac{1}{1 + \frac{\nu}{2} |\xi|^a} = \mathcal{LFG}(1, \xi).$$

Recall that

$$\mathcal{LFG}_H(s, \xi) = \frac{1}{s + |\xi|^2} \quad \text{and} \quad \mathcal{LFG}_W(s, \xi) = \frac{1}{s^2 + |\xi|^2}$$

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$$\frac{1}{1 + \frac{\nu}{2} |\xi|^a} = \mathcal{LFG}(1, \xi).$$

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$$\mathcal{LFG}_H(s, \xi) = \frac{1}{s + |\xi|^2} \quad \text{and} \quad \mathcal{LFG}_W(s, \xi) = \frac{1}{s^2 + |\xi|^2}$$

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Can we make it even sharper?

Indeed, with slightly more effort, one can show that the *reverse Cauchy-Schwarz inequality* above is not too bad:

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\int_0^\infty e^{-t} \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt \right) = \log \left(2^{2(b+r) - \frac{b\alpha}{a}} \rho_{\nu, a}(\gamma) \right)$$

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$$\mathcal{M}_{a,d}(\gamma, \theta) := \sup_{g \in \mathcal{F}_a} \left\{ \left\langle g^2 * g^2, \gamma \right\rangle_{L^2(\mathbb{R}^d)}^{1/2} - \frac{\theta}{2} \mathcal{E}_a(g, g) \right\}$$

► $\mathcal{E}_a(g, g) := (2\pi)^{-d} \int_{\mathbb{R}^d} |\xi|^a |\mathcal{F}g(\xi)|^2 d\xi$

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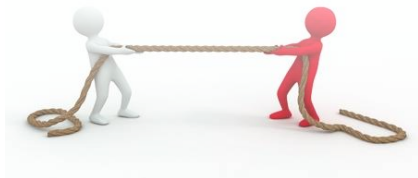
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$$-(-\Delta)^{a/2} u$$

$$\dot{W}u$$

Smoothing

Roughening



Upgrading from second moment to p -th moment...

1. Upper bound: hypercontractivity

$$\|u(t, x)\|_p \leq \|u_{(p-1)\theta}(t, x)\|_2, \quad \text{for all } p \geq 2, t > 0 \text{ and } x \in \mathbb{R}^d.$$

2. Lower bound: Construct/find the exact function that achieves the upper bound.

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More general SPDEs?

We have seen that all we need seem to be

$$\mathcal{LFG}(1, \xi) = \frac{1}{1 + \frac{\nu}{2} |\xi|^a} \quad (\star)$$

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Indeed, we do have a class of SPDEs,
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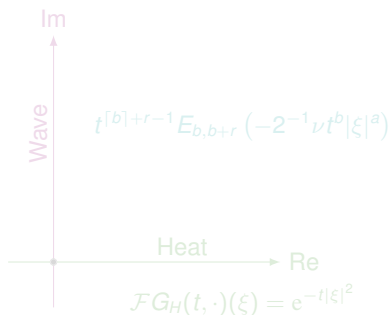
In C., Hu, Nualart 19, we have studied:

$$\begin{cases} \left(\partial_t^b + \frac{\nu}{2} (-\Delta)^{a/2} \right) u(t, x) = I_t^r \left[\sqrt{\theta} u(t, x) \dot{W}(x) \right] & x \in \mathbb{R}^d, t > 0, \\ u(0, \cdot) = 1 & b \in (0, 1], \\ u(0, \cdot) = 1, \quad \partial_t u(0, \cdot) = 0 & b \in (1, 2), \end{cases}$$

for $a \in (0, 2]$, $b \in (0, 2)$, $r \geq 0$, $\nu > 0$ and $\theta > 0$.

$$\left(\partial_t^b + (-\Delta)^{a/2}\right) u(t, x) = I_t^r \left[\sqrt{\theta} u(t, x) \dot{W}(x) \right]$$

$$\mathcal{F}G_W(t, \cdot)(\xi) = \frac{\sin(t|\xi|)}{|\xi|} = \text{Im} \frac{e^{it|\xi|}}{|\xi|}$$



$$\mathcal{L}\mathcal{F}G_W(\circ, \cdot)(s, \xi) = \frac{1}{s^2 + |\xi|^2}$$

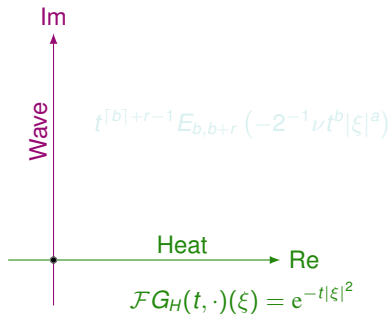
$$\frac{s^{-r}}{s^b + |\xi|^a}$$

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In particular, by taking $s = 1$, we have $\mathcal{L}\mathcal{F}G(1, \xi) = \frac{1}{1 + |\xi|^2}$

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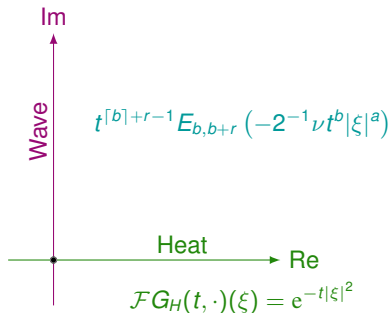
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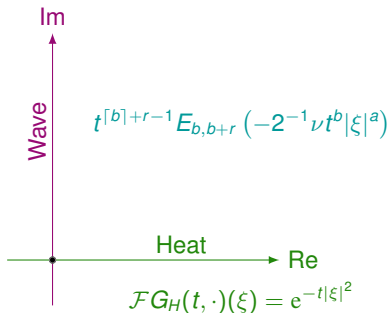
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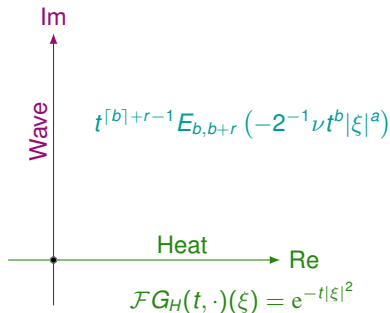
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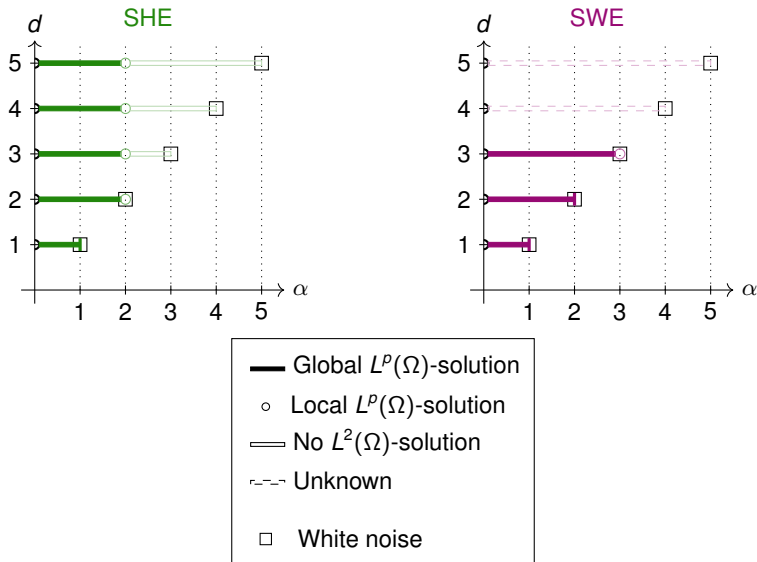
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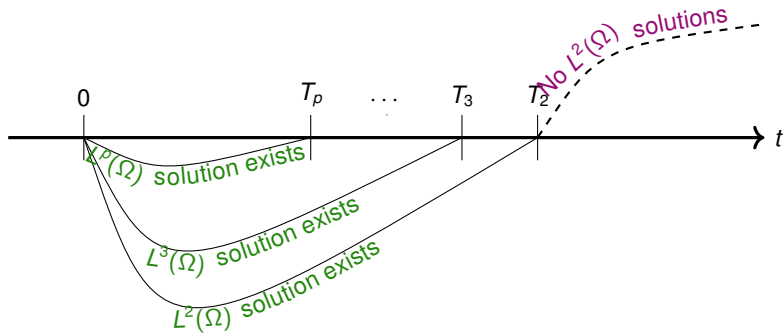
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Exact solvability ⁸



⁸Balan, L. Chen & X. Chen, AIHP 2021.

Exact phase transition in case of local solution



Theorem (Exact Asymptotics [Balan-C.-Chen 21])

(i) Under *Assumption A* with $0 < \alpha < d \leq 3$, by setting

$$t_p := (p-1)^{1/(4-\alpha)} t,$$

it holds that

$$\lim_{t_p \rightarrow \infty} t_p^{-\frac{4-\alpha}{3-\alpha}} \log \|u(t, x)\|_p = \theta^{\frac{1}{3-\alpha}} \left(\frac{1}{2}\right)^{\frac{\alpha}{2(3-\alpha)}} \frac{3-\alpha}{2} \left(\frac{2\mathcal{M}^{1/2}}{4-\alpha}\right)^{\frac{4-\alpha}{3-\alpha}}.$$

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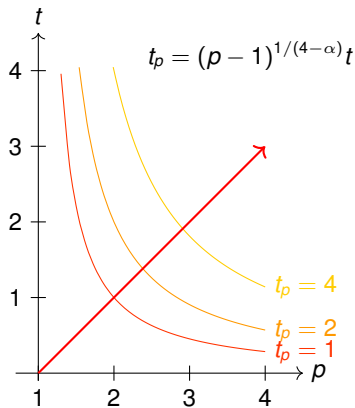
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Stochastic
Heat Equation

$$\lim_{t_p \rightarrow \infty} [(p-1)^{2/((4-\alpha))} t]^{-\frac{4-\alpha}{2-\alpha}} \\ \times \log \|u(t, x)\|_p \\ || \\ \theta^{\frac{2}{2-\alpha}} \frac{2-\alpha}{4} \left(\frac{4\mathcal{M}}{4-\alpha} \right)^{\frac{4-\alpha}{2-\alpha}}$$

X. Chen 17 ...

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X. Chen, L. C. and R. Balan 21

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X. Chen, L. C. and R. Balan 21



$$\begin{aligned}
 & \lim_{t_p \rightarrow \infty} \left[(p-1)^{1-1/\beta} t \right]^{-\frac{4-\alpha}{2-\alpha}} \\
 & \quad \times \log \|u(t, x)\|_p \\
 & \quad \parallel \\
 & \quad \left(\frac{1}{2} \right) \left(\frac{2a}{2a(b+r) - b\alpha} \right)^\beta \\
 & \quad \times \left(\theta \nu^{-\alpha/a} \mathcal{M}_a^{\frac{2a-\alpha}{a}} \right)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1 \right), \\
 & \quad \beta := \frac{2(b+r) - \frac{b\alpha}{a}}{2(b+r) - \frac{b\alpha}{a} - 1}
 \end{aligned}$$

This talk is based on:

1. C., Y. Hu and D. Nualart Nonlinear stochastic time-fractional slow and fast diffusion equations on $\mathbb{R}^d$. *Stochastic Process. Appl.*, 2019.
2. R. Balan, L. C., and X. Chen Stochastic wave equation with time-independent noise. *Ann. Inst. Henri Poincaré Probab. Stat.* in press, 2022
3. C. and N. Eisenberg Interpolating the stochastic heat and wave equations with time-independent Gaussian noise: solvability and exact asymptotics. *SPDE: Anal. & Computat.*, accepted pending revision, 2022

Other references

1. R. Balan and J. Song Second order Lyapunov exponents for the parabolic and hyperbolic Anderson models. *Bernoulli* 2019.
2. R. Bass, X. Chen, and M. Rosen Large deviations for Riesz potentials of additive processes. *Ann. Inst. Henri Poincaré: Probab. & Stat.* 2009.
3. Chen, X. Moment asymptotics for parabolic Anderson equation with fractional time-space noise: in Skorohod regime. *Ann. Inst. Henri Poincaré: Prob. Stat.* **53** 819–841, 2017.
4. Chen, X. Parabolic Anderson model with rough or critical Gaussian noise. *Ann. Inst. Henri Poincaré: Prob. Stat.*, 2019.
5. Chen, X., A. Deya, J. Song and S. Tindel Solving the hyperbolic Anderson model 1: Skorohod Setting. *Ann. of Prob.*, 2022.
6. Y. Hu Heat equation with fractional white noise potential. *Appl. Math. Optim.* 2001.
7. Y. Hu, J. Huang, D. Nualart, and S. Tindel Stochastic heat equations with general multiplicative Gaussian noises: Hölder continuity and intermittency. *Electr. J. Probab.* 2015.

Thank you for listening!

Le Chen (le.chen@auburn.edu)

Reverse Cauchy-Schwarz inequality

Lemma (Xia Chen ...)

If $f : [0, \infty) \rightarrow [0, \infty)$ is a non-decreasing function, then

$$2 \int_0^\infty e^{-2t} f^2(t) dt \leq \left(\int_0^\infty e^{-t} f(t) dt \right)^2.$$

Proof.

Let τ and τ' be two independent random variables with exponential distribution of mean 1. Then $\tau \wedge \tau'$ has an exponential distribution with density $g(t) = 2e^{-2t}$ and

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